Achmad Solaeman

Junior Lecturer · Internet Engineering Technology · Universitas Gadjah Mada

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EDUCATION

Master of Engineering (Information Technology)
Universitas Gadjah Mada · Yogyakarta, Indonesia · 2023 – 2025

- Specializing in deep learning, with a focus on computer vision and time series classification
- GPA: 3.80 / 4.00
- Relevant coursework: Artificial Intelligence, Big Data, Cloud Computing, Decision Support Systems

Bachelor of Applied Engineering (Internet Engineering Technology)
Universitas Gadjah Mada · Yogyakarta, Indonesia · 2013 – 2018

- Relevant coursework: Computer Networks, Network Security, Edge Computing

PROFESSIONAL EXPERIENCE

Junior Lecturer (Asisten Ahli), Internet Engineering Technology
Universitas Gadjah Mada · Sleman, Yogyakarta · 02/2026 – Present

- Teach courses and laboratory sessions in information security, databases, network design, and digital infrastructure (see *Teaching*)

Laboratory Assistant, UPT Komputer and UPT Bahasa
Universitas Muria Kudus · Kudus, Indonesia · 2020 – 01/2026

- Administered Microsoft Office Specialist and Microsoft Certified Educator certification exams for 250 candidates per semester, ensuring compliance with testing protocols
- Proctored British Council EnglishScore certification exams for 300 test-takers annually, maintaining examination integrity
- Developed a computerized English proficiency testing system and digitized certificate management, improving administrative efficiency

English & Arabic Language Tutor
Pondok Pesantren Sulaimaniyah · Pati, Indonesia · 2021 – 2024

- Taught English and Arabic across 5 grade levels (junior to senior high school), serving 80 students weekly

Teaching Assistant
Universitas Gadjah Mada · Yogyakarta, Indonesia · 2015 – 2017

- Led laboratory sessions for Computer Networking, Network Security, Algorithms & Data Structures, Computer Applications, and Fundamentals of Computer Engineering

Intern, Fault Management
PT Indonesia Comnets Plus (ICON+) · Depok, Indonesia · 02/2017 – 05/2017

- Monitored network infrastructure in real time through the Network Operation Center (NOC), identifying and documenting system faults

TEACHING

2026/2027

Odd semester (current) · Junior Lecturer · Universitas Gadjah Mada

- Database Systems (Basis Data)
- Digital Infrastructure Support Laboratory Work

2025/2026

Even semester · Junior Lecturer · Universitas Gadjah Mada

- Information Security 2
- Occupational Health and Safety (K3)
- Network Design Laboratory Work
- Database Laboratory Work

2015–2017

Teaching Assistant · Universitas Gadjah Mada

- Computer Networking
- Network Security
- Algorithms & Data Structures
- Computer Applications
- Fundamentals of Computer Engineering

PUBLICATIONS

2025

A Systematic Literature Review of Transfer Learning for Eye Tracking

Achmad Solaeman, Syukron Abu Ishaq Alfarozi, Sunu Wibirama

ICADEIS 2025 · Conference paper · IEEE · pp. 1-7 · doi:10.1109/ICADEIS65852.2025.10933367

A review of 29 studies (2018–2024) showing how transfer learning addresses data scarcity and individual variability in gaze classification and estimation, with best practices for preprocessing and fine-tuning.

2024

Robust Fetal Head Circumference Calculation on Ultrasound Images Using Just-in-Time Compiled Ellipse Fitting

Pandega Abyan Zumarsyah, **Achmad Solaeman**, Fhareza Alvindo, Hanung Adi Nugroho

TENCON 2024 · Conference paper · IEEE · pp. 161-164 · doi:10.1109/TENCON61640.2024.10903012

A fast, memory-efficient method based on JIT-compiled Randomized Hough Transform ellipse fitting, reaching Dice 0.93 and Hausdorff 3.88 mm in 0.41 s per image with under 800 kB of memory.

Personalized medicine meets artificial intelligence: beyond “hype”, towards the metaverse

Eko Wahyudi, Nur Eko Windianto, Gheri Febri Ananda, **Achmad Solaeman**, Wisnu Setiawan

Informatics for Health and Social Care · Book review · Taylor & Francis · pp. 149-152 · doi:10.1080/17538157.2024.2347230

A review of a book on how AI and personalized medicine are shaping digital health, data-driven care, and patient outcomes.

SERVICES

Program Committee
2026

Technical Program Committee Member

DELCON 2026, BITS Pilani · Conference

Peer Review
Ongoing

Reviewer

Journal of Internet and Software Engineering (JISE) · Journal

Community Service
2026

Community Service Team — IoT-based Smart School Learning for Vocational High School Students

SMK Negeri 2 Depok, Sleman · with PT Lifemedia Yogyakarta · PkM

A community service program by the Department of Electrical Engineering and Informatics (DTEDI), Vocational College UGM, delivering an IoT Smart School training module aligned with Kurikulum Merdeka, a lab-scale trainer kit, and a project-based workshop for students.

PROJECTS

A Hybrid Deep Learning Approach for Accurate Eye Movement Classification
Master's thesis · 02/2025 – 06/2025

A lightweight hybrid deep learning model for near real-time eye-movement classification, achieving over 91% F1-score on the GazeCom dataset with only a 20 ms input window, enabling low-latency gaze interaction for assistive technologies.

ACL and Dynamic Control List in Software-Defined Networks with Floodlight Controller
Bachelor's final project · 09/2017 – 11/2017

A desktop application to manage outbound packet flow via Access Control Lists, extended with a Dynamic Control List that builds firewall rules automatically in collaboration with a honeypot.

CERTIFICATES

- IT Specialist – Artificial Intelligence · Certiport · ID 422dd365-f73a-4211-ab5d-9ac0e0d7d0bc
- IT Specialist – Data Analytics · Certiport · ID 9282bd0b-43e6-4d67-b17e-4967f0547211
- IT Specialist – Cybersecurity · Certiport · ID f57f3d86-d556-4ace-96b6-f66a46167193
- IT Specialist – Networking · Certiport · ID 4f7038dc-3651-451f-b23b-59bdd273eddd
- Microsoft Certified: Azure Data Fundamentals · Microsoft · ID 7d6664e0-016c-4180-a085-ac69ff8fc11a
- Microsoft Office Specialist: Expert (Office 2019) · Microsoft · ID 97bd0866-52fd-4ffa-a939-5cc40567b63e
- Microsoft Certified Educator (MCE) · Microsoft · ID 05334202-4373-486f-a87f-a860ac0aeddc
- PMI Project Management Ready™ · PMI · ID 67f52562-bd65-44bf-a791-1d19c840951f

SKILLS

- **Machine Learning:** Python, PyTorch, TensorFlow
- **Databases:** MySQL, PostgreSQL, MongoDB, database design
- **Web Development:** Laravel (PHP), Django (Python), React.js, WordPress
- **Cloud Computing:** Amazon Web Services (EC2, S3, Lambda)
- **Networking & Security:** network design, SDN (Floodlight), network monitoring (NOC)
- **Languages:** Indonesian (native), English, Arabic

COURSES & TRAINING

Artificial Intelligence & Big Data — Digital Talent Scholarship
Ministry of Communication and Information Technology, Republic of Indonesia · Yogyakarta · 10/2018 – 12/2018

Foundational and practical training in data preprocessing, machine learning models, and real-world applications.